

Disentangling high-order mechanisms and high-order behaviours in complex systems

To the Editor — Battiston et al¹, provide a comprehensive overview of how investigations of complex systems should take into account interactions between more than two elements, which can be modelled by hypergraphs and studied via topological data analysis. Following a separate line of enquiry, a broad literature has developed information-theoretic tools to characterize high-order interdependencies from observed data. While these could seem to be competing approaches aiming to address the same question, in this correspondence we clarify that this is not the case, and that a complete account of higher-order phenomena needs to embrace both.

The approaches reviewed by Battiston and colleagues put a special focus on what could be described as high-order mechanisms, which are usually modelled via Hamiltonians or dynamical laws involving beyond-pairwise interactions. A distinct, but complementary perspective is to focus on the resulting high-order behaviour, that is, patterns of activity that can be explained in terms of the whole but not the parts. Put simply, high-order mechanisms refer to the modelling of the data-generating process, while high-order behaviours refer to emergent properties within the resulting multivariate statistics. For example, in a spin glass system, high-order mechanisms would correspond to high-order terms in the system's Hamiltonian, while high-order behaviours would refer to the emergent patterns resulting from the corresponding Boltzmann distribution (Fig. 1a).

The investigation of high-order behaviour in data has a long history, stemming from foundational work in information theory² and its early applications in biophysics^{3,4}. These early efforts led to the formalization of high-order behaviour into the framework of partial information decomposition and its subsequent developments^{5–7}. This literature has established formal bases for the analysis of high-order interdependencies exhibited by groups of three or more variables, and a description of their (synergistic or redundant) nature. While our primary focus here is on information-theoretic techniques, a number of other frameworks exist for probing the dynamics of complex systems beyond pairwise interactions — for a comprehensive review, see ref. ⁸.

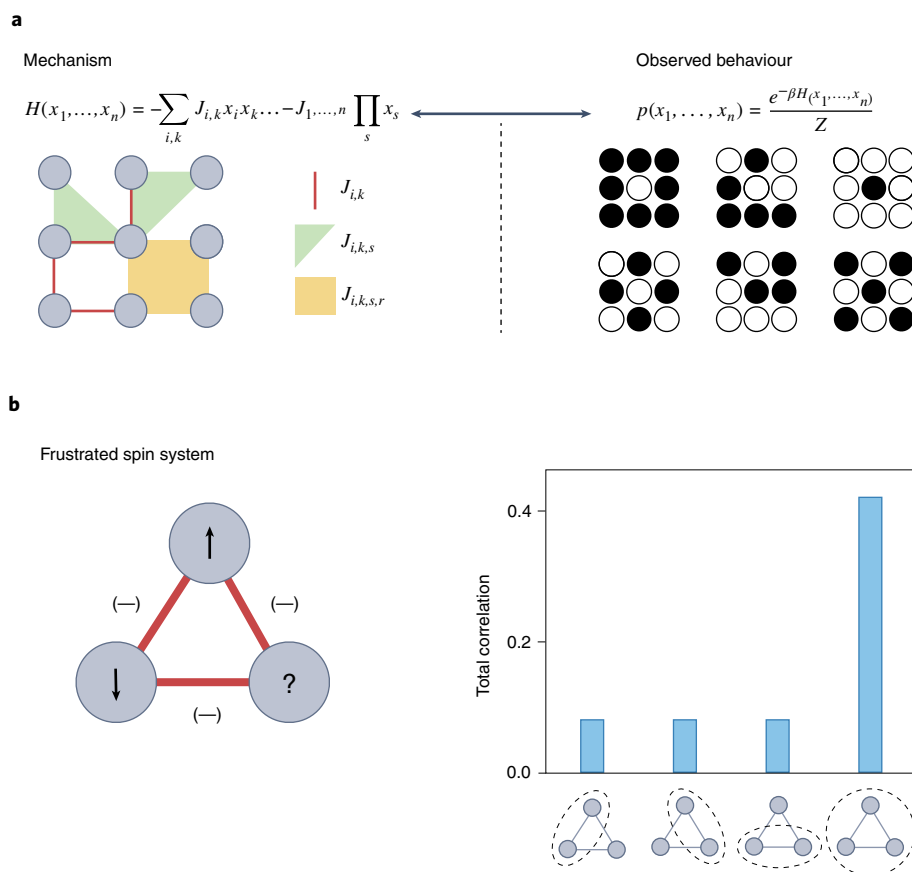


Fig. 1 | Contrasting high-order mechanisms and behaviours in complex systems. a, Complementarity of mechanisms and behaviours in Ising spin systems. The mechanism of a spin glass model is determined by a Hamiltonian H that accounts for pairwise and higher-order interactions, forming a hypergraph. The behaviour of the system is instead given by the frequencies of observed patterns, which can manifest as high-order interdependencies such as synergy or redundancy^{5–7}. **b**, Dissociation of low-order mechanisms and high-order behaviour in frustrated spin systems. A small frustrated spin model, wherein spins cannot satisfy their tendency to differ from their neighbours¹⁰. The total interdependency (measured by total correlation³) significantly exceeds the sum of the three pairwise interdependencies, illustrating how low-order mechanisms can give rise to higher-order behaviour that cannot be explained from pairwise statistics.

Mechanisms and behaviours address fundamentally different questions: the former address how the system is structured, while the latter focus on emergent properties related to what the system ‘does’. Crucially, these questions are not interchangeable — intuition might suggest that high-order behaviour necessarily rests upon high-order mechanisms, but this is not the case (Fig. 1b). Therefore, neglecting high-order behaviour risks missing important aspects of complex

phenomena. Furthermore, when the goal is to identify high-order behaviour, high-order methods relying only on pairwise statistics (for example, simplicial complexes built from a correlation matrix) may be in principle insufficient as significant information can be present only in the joint probability distribution and not in the pairwise marginals (Fig. 1b). There is a great need for inferential tools that can connect these two worlds — for example, model

selection methods to identify higher-order mechanisms that best match observed high-order behaviour, extending prior work on pairwise networks⁹.

In summary, differentiating between high-order mechanisms and behaviours allows greater precision in articulating hypotheses, and in choosing the right tools to probe them. This distinction will facilitate the collaboration between different communities devoted to the study of complex systems, as the complementarity of these perspectives will provide a more powerful and encompassing avenue for deepening our understanding of high-order phenomena. 

Fernando E. Rosas ^{1,2,3,4,19} ,
Pedro A. M. Mediano ^{5,6,19} ,
Andrea I. Luppi ^{7,8,9,10,19} ,
Thomas F. Varley ^{11,12}, **Joseph T. Lizier** ¹³,
Sebastiano Stramaglia^{14,15},
Henrik J. Jensen ^{3,16,17} and
Daniele Marinazzo¹⁸

¹Data Science Institute, Imperial College London, London, UK. ²Centre for Psychedelic Research, Imperial College London, London, UK. ³Centre for Complexity Science, Imperial College London,

London, UK. ⁴Department of Informatics, University of Sussex, Brighton, UK. ⁵Department of Psychology, University of Cambridge, Cambridge, UK.

⁶Department of Psychology, Queen Mary University of London, London, UK. ⁷University Division of Anaesthesia, University of Cambridge, Cambridge, UK. ⁸Department of Clinical Neurosciences, University of Cambridge, Cambridge, UK.

⁹Leverhulme Centre for the Future of Intelligence, University of Cambridge, Cambridge, UK. ¹⁰The Alan Turing Institute, London, UK. ¹¹Department of Psychological & Brain Sciences, Indiana University, Bloomington, IN, USA. ¹²School of Informatics, Computing, and Engineering, Indiana University, Bloomington, IN, USA. ¹³Centre for Complex Systems and Faculty of Engineering, The University of Sydney, Sydney, New South Wales, Australia.

¹⁴Dipartimento Interateneo di Fisica, Università degli Studi di Bari Aldo Moro, Bari, Italy. ¹⁵INFN, Sezione di Bari, Bari, Italy. ¹⁶Department of Mathematics, Imperial College London, London, UK. ¹⁷Institute of Innovative Research, Tokyo Institute of Technology, Tokyo, Japan. ¹⁸Department of Data Analysis, Ghent University, Ghent, Belgium. ¹⁹These authors contributed equally: Fernando E. Rosas, Pedro A. M. Mediano, Andrea I. Luppi.

e-mail: f.rosas@imperial.ac.uk; pam83@cam.ac.uk; al857@cam.ac.uk

Published online: 21 March 2022

<https://doi.org/10.1038/s41567-022-01548-5>

References

1. Battiston, F. et al. *Nat. Phys.* **17**, 1093–1098 (2021).
2. Watanabe, S. *IBM J. Res. Dev.* **4**, 66–82 (1960).
3. Gat, I. & Tishby, N. in *Adv. Neural Inf. Process. Syst.* (eds Kearns, M. J. et al.) 111–117 (MIT Press, 1999).
4. Tononi, G., Sporns, O. & Edelman, G. M. *Proc. Natl Acad. Sci. USA* **91**, 5033–5037 (1994).
5. Williams, P. L. & Beer, R. D. Preprint at <https://arxiv.org/abs/1004.2515> (2010).
6. Lizier, J. T., Bertschinger, N., Jost, J. & Wibral, M. *Entropy* **20**, 307 (2018).
7. Mediano, P. A. et al. Preprint at <https://arxiv.org/abs/2109.13186> (2021).
8. Battiston, F. et al. *Phys. Rep.* **874**, 1–92 (2020).
9. Peixoto, T. P. *Phys. Rev. Lett.* **123**, 128301 (2019).
10. Matsuda, H. *Phys. Rev. E* **62**, 3096 (2000).

Acknowledgements

F.R. was supported by the Ad Astra Chandaria Foundation. P.M. was funded by the Wellcome Trust (grant no. 210920/Z/18/Z). A.L. was supported by a Gates Cambridge Scholarship (OPP1144). T.V. is supported by the NSF-NRT grant 1735095, Interdisciplinary Training in Complex Networks and Systems at Indiana University Bloomington. S.S. was supported by the Italian MIUR PRIN 2017 (2017WZFTZP). A.L. and T.V. gratefully acknowledge the support of a Young Researchers' Bridge Grant from the Complex Systems Society.

Competing interests

The authors declare no competing interests.